

Annotation Guidelines for Yoruba–German Code-Switched Sentences

The annotation process consists of two phases:

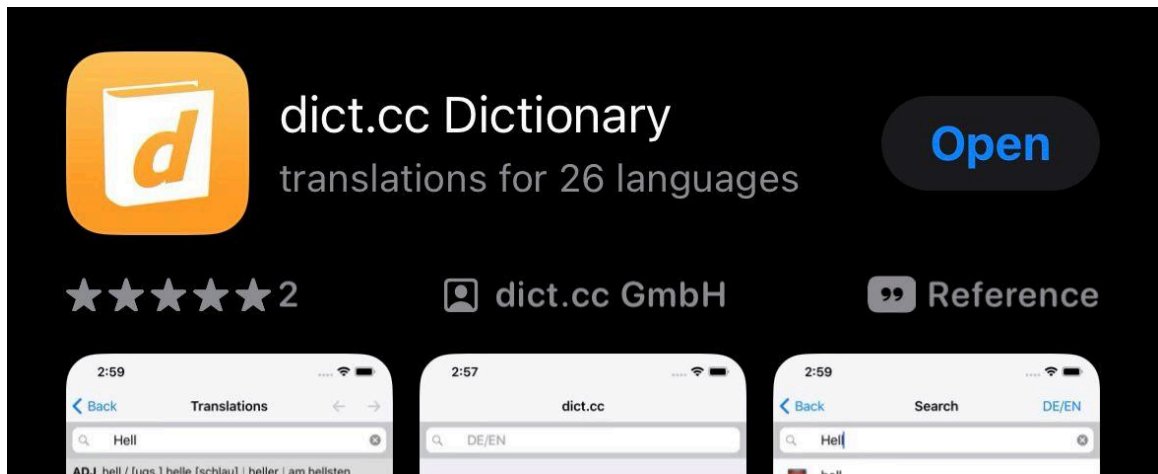
Phase 1: Correction of Code-Switched Sentences

Objective

Refine the generated code-switched sentences to ensure they reflect natural, grammatically coherent, and culturally appropriate Yorùbá–German code-switching patterns.

Instructions

- **Authenticity:** Modify sentences where necessary to reflect natural code-switching patterns commonly used by bilingual speakers of Yorùbá and German.
- **Cultural Relevance:** Ensure that corrected sentences align with realistic cultural and social contexts relevant to the language pair.
- **Structural Flexibility:** Annotators may restructure sentences when necessary to improve fluency, grammaticality, or natural code-switching behaviour.
- **Tonal Language Consideration:** Since Yorùbá is a tonal language, tonal diacritics are optional during this phase. However, under-dots should be preserved where applicable.
- **Preservation of Well-Formed Sentences:** If a sentence is already natural, grammatically correct, and contextually appropriate, no modification should be made.
- **Handling German Embedded Words:** Annotators may encounter unfamiliar German words or expressions. In such cases, annotators are encouraged to consult reliable German dictionaries to verify the meaning and contextual appropriateness of the embedded German words.



- **Commenting on Corrections:** Use the linguistic observation column to document observations about the LLM-generated sentence, including issues such as unnatural code-switching patterns, semantic inconsistencies, grammatical errors, limited lexical diversity, culturally inappropriate expressions, or other notable characteristics of the model's output.
- **No Changes Made:** If no correction is required, leave the linguistic observation column blank.
- **Code-Switching Requirement:** Ensure that every corrected sentence contains actual Yorùbá–German code-switching. Fully monolingual sentences should be flagged for review.
- **Semantic Consistency:** Corrections should preserve the original meaning and intent of the sentence as much as possible.

Each domain dataset contains 500 generated code-switched sentences. Two annotators will participate in the annotation process for each domain.

- Annotator A will annotate the first 250 sentences.
 - Annotator B will annotate the remaining 250 sentences.
 - Annotators may restructure sentences where necessary to improve naturalness, fluency, grammaticality, and semantic coherence.
 - The corrected sentences should maintain Yorùbá as the matrix language and German as the embedded language.
 - Annotators should ensure that all corrected sentences contain meaningful and natural code-switching patterns.
- **Naturalness:** Avoid overly literal or artificial mixing of languages. The final sentence should sound natural to bilingual speakers.

Recommended Annotation Columns

Column Name	Purpose
Author	Unique sentence identifier
domain	Dataset domain, such as business, education, or food
generated_sentence	Original LLM-generated code-switched sentence
corrected_sentence	Human-corrected sentence
english_translation	English translation of the corrected sentence
linguistic_observation	Notes about grammar, switching patterns, or semantic issues
validation_status	Accepted or rejected

quality_flag	Indicates whether a sentence is monolingual, repetitive, good, or requires annotation/revision. This column helps annotators and validators identify the overall quality and usability of generated code-switched sentences.
--------------	--

Phase 2: Cross Validation

After the first annotation phase, the annotators will exchange their assigned sentence sets for validation and quality control.

- Annotator A will validate the 250 sentences originally annotated by Annotator B.
- Annotator B will validate the 250 sentences originally annotated by Annotator A.
- Validators will review sentence quality, grammatical correctness, semantic meaning, and naturalness of code-switching.
- Validators will use the validation_status column to indicate the final decision for each sentence.

Validation Status Labels

Validation Status	Meaning
Accepted	The sentence is natural, meaningful, grammatically coherent, and requires no further correction.
Edited	The sentence is understandable but contains minor grammatical, semantic, or structural errors that were corrected during validation.
Rejected	The sentence lacks clear meaning, contains unnatural code-switching, or is grammatically incoherent.